```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\DELL\Downloads\cn -12> javac UDPEchoServer.java
PS C:\Users\DELL\Downloads\cn -12> java UDPEchoServer
UDP Echo Server listening on port 6000
Received from /127.0.0.1:65524 -> hi
Received from /127.0.0.1:65524 -> dai enna pandra
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\DELL\Downloads\cn -12> javac ChatClient.java
PS C:\Users\DELL\Downloads\cn -12> java ChatClient
Enter your name:
nithish
Welcome, nithish! Type /quit to exit.
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

        at java.base/java.net.ServerSocket.<init>(ServerSocket.java:138)
        at ChatServer.main(ChatServer.java:12)
PS C:\Users\DELL\Downloads\cn -12> netstat -ano | findstr :5000
  TCP    0.0.0.0:5000           0.0.0.0:0              LISTENING       10340
  TCP    127.0.0.1:5000         127.0.0.1:50503        ESTABLISHED     10340
  TCP    127.0.0.1:50503        127.0.0.1:5000         ESTABLISHED     2852
  TCP    [::]:5000              [::]:0                 LISTENING       10340
PS C:\Users\DELL\Downloads\cn -12> taskkill /PID 10340 /F
SUCCESS: The process with PID 10340 has been terminated.
PS C:\Users\DELL\Downloads\cn -12> java ChatServer
Chat server started on port 5000
nithish connected from /127.0.0.1
```

No: 12(a)

Echo client server using Tcp Sockets

AIM:-

To write and execute a c/ java socked program to implement a TCP Echo client and echo Server.

ALGORITHM:-

Server Side :

1- Create a Socket using socket ().

2- Bind the socket the IP address and port Number using bind()

3. listen for incoming client connecting using listent()

4. accept the connection request from the client using accept()

5. Read / Receiver the message send by the client.

6. Send / Echo the same message back to the client

7- Close the client and server sockets -

PROGRAM CODE:-

Echo Server.java

```
import java.i.o*;
import java.net.*;
public class EchoServer {
 public static Void main(String[] args) {
try {
ServerSocket ServerSocket = new ServSocket(5000);
System.out.println("Echo server started waiting for client");
Socket socket = ServerSocket.accept();
System Out.println("Client Connected!");
BufferReader in = New BufferReader(new Input stream
                Reader(Socket.get Input stream()));
print Writer out = new print Writer( Socket get output
                                    Stream(),true);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\DELL\Downloads\cn -12> javac TCPEchoServer.java
PS C:\Users\DELL\Downloads\cn -12> java TCPEchoServer
Error: Could not find or load main class TCPEchoServer
Caused by: java.lang.ClassNotFoundException: TCPEchoServer
PS C:\Users\DELL\Downloads\cn -12> javac TCPEchoServer.java
PS C:\Users\DELL\Downloads\cn -12> java TCPEchoServer
TCP Echo Server listening on port 5000
Client connected: /127.0.0.1:50503
Received: hi
Received: hi i am nithish
PS C:\Users\DELL\Downloads\cn -12>
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\DELL\Downloads\cn -12> javac UDPEchoClient.java
PS C:\Users\DELL\Downloads\cn -12> java UDPEchoClient
UDP Echo Client ready to send to localhost:6000
Type messages to send (type 'bye' to quit):
hi
Server: Echo: hi
dai enna pandra
Server: Echo: dai enna pandra
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\DELL\Downloads\cn -12> javac TCPEchoClient.java
PS C:\Users\DEL1\Downloads\cn -12> java TCPEchoClient
Connected to TCP Echo Server at localhost:5000
Type messages to send (type 'bye' to quit):
hi
Server: Echo: hi
hi i am nithish
Server: Echo: hi i am nithish
```

```
String inputline = in.readline();
System.out.println = in.readline();
Socket.close()
Secure socked.close(),
} catch (Io Exceptiona){
        e.printstackTower);
  }
}
}
```

Echo client.java

```
import java.io.*;
import java.net.*;
public class Echo Client{
  public static Void main(String[] args){
    try{
      Socket socket = new socket("localhost", 5000);
      System.out.println("connect to Echo server!");
  BufferReader user Input = new BufferReader(New
                    Input stream Reader (System.in);
  BufferReader in = new BufferReader new Input stream Reader
        (Socket.get Input stream ()));
  System.out.print("Enter message to Echo");
  String message = user Input.readline();
  Out.println(message);
  String echo Response = in.readline;
System.out.println("Server echo Respons:"+ echoresponse)
Socket.close();
}catch(Io Exceptione){
    e.print stack trace();
  }
 }
}
```

Server output:

Echo secue started waiting for client

Client connected

Received from client : Hello computer Network lab

Echoed back to client

Client Output:

Connected to Echo server: -

Enter message to Echo : Hello computer Network lab

Server echo Response : Hello computer Network lab

Result :-

The TCP echo client / Server program was successfully implemented, compiled, and executed and the Server correctly echoed the received message back to the client.